Jangara: A Stablecoin Banking Solution for Global Financial Integration

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Abstract

This paper presents Jangara, a banking solution initially designed for economies experiencing hyperinflation and currency instability, but with applications across the global financial ecosystem. By leveraging stablecoin technology pegged to the US dollar, Jangara addresses the scarcity of physical USD while enabling efficient local transactions, fractional payments, and financial inclusion. The system operates without charging user fees, deriving revenue solely from the appreciation of USD against local currencies and arbitrage opportunities in currency fluctuations. Through mathematical modeling and comparative analysis, we demonstrate that this approach represents a viable financial innovation with significant potential benefits for users in both distressed and stable economies. Additionally, Jangara creates a global financial network that optimizes remittance flows, currency exchange, and access to foreign currency for businesses that need it most.

1 Introduction

In economies suffering from hyperinflation and monetary instability, citizens often resort to using foreign currencies, particularly the US dollar, as a store of value and medium of exchange (Mundell, 1961; Calvo and Reinhart, 2002). However, this dollarization presents significant challenges: physical dollars are scarce, cannot be easily divided for small transactions, and acquiring them requires substantial effort not reflected in their nominal value (Kokenyne et al., 2010).

Recent data indicates that in Zimbabwe, approximately 70% of transactions are conducted in US dollars despite the reintroduction of the Zimbabwean dollar (Chikumbu, 2021). Similarly, Venezuela has seen over 60% of retail transactions conducted in dollars despite the official currency being the bolivar (Perez, 2020).

However, the remittance ecosystem that supplies these dollars is itself highly inefficient. Each year, over \$550 billion in remittances flow to developing countries, with traditional providers charging an average of 7% in fees (World Bank, 2021). Once received, these dollars face another inefficiency: they often circulate among everyday consumers and retailers who struggle with denominations and change, while importers and suppliers who genuinely need foreign currency for international purchases struggle to accumulate sufficient amounts.

2 Problem Statement

The current financial system presents multiple inefficiencies across both distressed and stable economies:

2.1 In Distressed Economies

- Scarcity of physical US dollars
- Difficulty in making fractional payments (lack of coins)
- High transaction costs in acquiring US dollars
- Limited financial infrastructure
- Mismatch between those who possess USD and those who need it for imports

The magnitude of these challenges can be quantified. For instance, the premium paid to acquire physical US dollars in Zimbabwe has been reported to reach up to 40% in informal markets (Makochekanwa, 2018):

$$\text{Dollar Premium} = \frac{\text{Local Market USD Rate} - \text{Official USD Rate}}{\text{Official USD Rate}} \times 100\% \tag{1}$$

2.2 In Stable Economies

- High costs for sending remittances to family members in distressed economies

- Inefficient currency exchange with significant spreads and limited availability
- Lack of transparency in cross-border transactions
- Limited working hours for traditional foreign exchange services

3 Proposed Solution: The Jangara Banking System

Jangara offers a stablecoin-based banking solution that functions as both a local financial infrastructure and a global cross-border network. The system provides:

3.1 Digital Dollar Equivalent

A stablecoin pegged 1:1 with the US dollar, allowing users to transact without needing physical dollars for local commerce. The stability mechanism follows the collateralized model described by Bullmann et al. (2019):

$$V_{Jangara} = \frac{R_{USD}}{C_{ratio}} \tag{2}$$

Where $V_{Jangara}$ represents the total value of Jangara tokens in circulation, R_{USD} is the USD reserve holdings, and C_{ratio} is the collateralization ratio (ideally maintained at 1:1).

3.2 Fractional Transactions

Unlike physical dollars, where coins are typically unavailable, the digital nature of the stablecoin allows for infinitely divisible transactions down to the cent level. This divisibility can be modeled as:

$$T_{min} = \frac{1}{10^d} \tag{3}$$

Where T_{min} is the minimum transaction size and d is the number of decimal places supported (typically 2 for dollars, but potentially higher in the digital implementation).

3.3 Fee-Free Banking

The system operates without charging transaction fees to users, removing financial barriers to adoption. This zero-fee model is sustainable because revenue is derived solely from currency fluctuation opportunities rather than user fees.

3.4 Global Network Effects

The value of the Jangara network increases with each additional country and currency added to the system. This can be expressed using Metcalfe's Law adapted for currency networks:

$$V_{network} \propto n(n-1)/2 \tag{4}$$

Where n is the number of currencies in the network. Each new currency added creates not just one new potential exchange route but $n-1$ new routes.

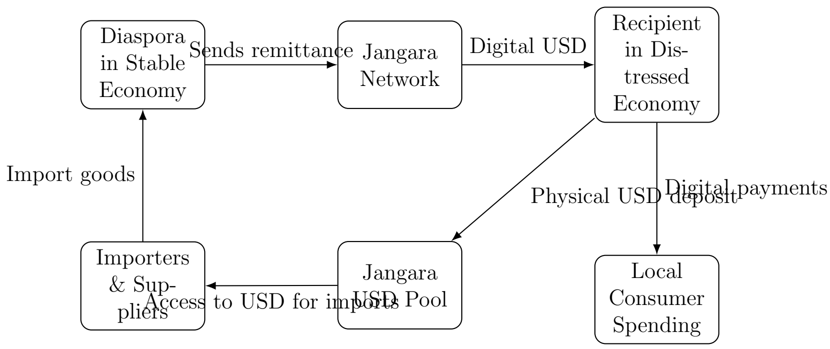

Figure 1: The Jangara Ecosystem: Optimizing Currency Flows

4 Revenue Model

Instead of charging fees, Jangara generates revenue through:

4.1 Currency Fluctuation Arbitrage

By operating across multiple currency markets simultaneously, Jangara can capitalize on small price differences between markets:

$$R_{arb} = \sum_{i=1}^{n} \sum_{j=1}^{n} V_{i,j} \times |P_{i,j}^{A} - P_{i,j}^{B}| \tag{5}$$

Where R_{arb} is arbitrage revenue, $V_{i,j}$ is the volume exchanged between currencies i and j, and $P^A_{i,j}$ and $P^B_{i,j}$ are the exchange rates in different markets.

4.2 Currency Depreciation Capture

In distressed economies, Jangara benefits from the appreciation of USD against local currencies:

$$R_{annual} = R_{USD} \times (L_{depreciation} - i_{USD} - C_{operational}) \tag{6}$$

Where R_{annual} is the annual revenue, R_{USD} is the total USD reserve, $L_{depreciation}$ is the rate of local currency depreciation against USD, i_{USD} is the interest rate paid on USD holdings, and $C_{operational}$ is the operational cost ratio.

4.3 Spread Optimization

By offering better rates than traditional providers while maintaining a small spread, Jangara can generate revenue while still saving users money:

$$R_{spread} = V_{exchange} \times S_{jangara} \tag{7}$$

Where R_{spread} is spread revenue, $V_{exchange}$ is exchange volume, and $S_{jangara}$ is Jangara's spread (typically 0.5-1% compared to traditional providers' 2-10%).

It's important to note that Jangara never risks users' principal deposits. Revenue is generated solely from the opportunities created by currency movements and network effects.

Table 1: Estimated Annual Revenue in Different Scenarios

Parameter	**Conservative**	**Moderate**	**Optimistic**
Reserve Size (USD)	\$10M	\$50M	\$100M
Local Currency Depreciation	30%	50%	70%
USD Interest Rate	2%	2%	2%
Operational Costs	10%	8%	6%
Exchange Volume	\$50M	\$200M	\$500M
Average Spread	0.5%	0.75%	1%
Net Annual Revenue	\$2.05M	\$21.5M	\$67M

5 Jangara in Stable Currency Economies

While Jangara was initially designed for distressed economies, the system provides substantial benefits in stable economies as well:

5.1 Foreign Exchange Platform

In stable economies, Jangara serves as an efficient foreign exchange platform with several advantages over traditional services:

- 24/7 availability versus limited banking hours
- Lower spreads (typically 0.5-1% versus 2-10%)
- No minimum transaction amounts
- Instant settlement versus traditional 1-3 day clearance periods

Table 2: Foreign Exchange Cost Comparison

Service	**Average Spread**	**Transaction Time**	**Additional Fees**
Traditional Banks	5-10%	1-3 days	Yes
Forex Bureaus	2-5%	Same day	Sometimes
Credit Cards	2.5-3%	Instant	Yes
Traditional Remittance	7-10%	1-2 days	Yes
Jangara	0.5-1%	Instant	No

5.2 Remittance Optimization

For diaspora communities in stable economies, Jangara provides a superior remittance channel:

- Zero fees compared to traditional remittance fees (5-10%)
- Instant transfers versus 1-3 day waiting periods
- Recipients receive digitally divisible currency, solving the "change problem"
- Transparency in exchange rates and transaction status

The remittance cost savings can be calculated as:

$$S_{remittance} = V_{remittance} \times (F_{traditional} - F_{jangara}) \tag{8}$$

Where $S_{remittance}$ is total savings, $V_{remittance}$ is remittance volume, $F_{traditional}$ is the average fee percentage for traditional services (typically 7%), and $F_{jangara}$ is Jangara's fee (0%).

5.3 Foreign Currency Access for Businesses

For businesses in stable economies that engage in international trade, Jangara provides:

- Direct access to pooled foreign currency reserves
- Lower exchange costs for international payments
- Hedging opportunities against currency fluctuations
- Streamlined cross-border payment processes

6 Implementation Challenges

- Regulatory considerations across multiple jurisdictions (Zetzsche et al., 2020)
- Technology infrastructure in developing economies
- Trust-building with potential users
- Security and fraud prevention
- Managing liquidity across multiple currency pools

Table 3: Comparison of Financial Solutions for Global Currency Exchange

Feature	**Physical USD**	**Local Currency**	**Cryptocurrency**	**Jang**
Stability	High	Variable	Low	Hig
Accessibility	Low	High	Medium	Hig
Divisibility	Low	High	High	Hig
Transaction Costs	High	Medium	High	Lo
Regulatory Compliance	Medium	High	Low	Hig
Infrastructure Requirements	Low	Low	High	Medi
Cross-Border Efficiency	Low	Low	Medium	Hig
Vendor USD Access	Low	N/A	Medium	Hig

7 Potential Impact

The Jangara system has the potential to:

7.1 In Distressed Economies

- Improve financial inclusion
- Reduce transaction costs for everyday commerce
- Provide stability in financial planning
- Redirect hard currency to where it creates maximum value (imports)
- Eliminate pricing inefficiencies caused by lack of fractional currency

7.2 In Stable Economies

- Reduce remittance costs for diaspora communities
- Provide more efficient foreign exchange services
- Facilitate international commerce with lower friction
- Create more integrated global currency markets

Table 4: Projected Economic Impact in Target Economies

Metric	Year 1	Year 3	Year 5
User Base (thousands)	50-100	300-500	1000-2000
Transaction Volume (USD millions)	5-10	50-100	200-500
Cost Savings for Users (%)	10-15	15-25	20-30
Financial Inclusion Improvement (%)	2-5	10-15	20-30
Remittance Cost Savings (USD millions)	0.35-0.7	3.5-7	14-35
Vendor Foreign Currency Access Improvement (%)	5-10	20-30	40-60

The economic value generated can be modeled as:

$$E_{value} = T_{volume} \times S_{rate} + F_{inclusion} \times P_{gdp} + R_{volume} \times R_{savings} \tag{9}$$

Where E_{value} is the economic value generated, T_{volume} is the transaction volume, S_{rate} is the average savings rate on transactions, $F_{inclusion}$ is the financial inclusion improvement factor, P_{gdp} is the per capita GDP increase from improved financial access (Sahay et al., 2015), R_{volume} is remittance volume, and $R_{savings}$ is the percentage savings on remittances.

8 Closed-Loop Efficiency

The true innovation of Jangara is creating a closed-loop system that optimizes currency flows. In the current system:

1. Diaspora sends money via traditional remittance providers at high cost
2. Recipients receive physical USD which is difficult to use for small transactions
3. Local retailers struggle with providing change for USD transactions
4. Importers and suppliers who need USD for international purchases struggle to accumulate it

Jangara transforms this into:

1. Diaspora sends money through Jangara at zero cost
2. Recipients receive digital USD equivalent with perfect divisibility

3. Local transactions occur digitally with exact amounts
4. Physical USD is pooled and made available to importers/suppliers who genuinely need it

This optimization creates value across the entire ecosystem without charging fees to any participant. The efficiency gain can be quantified as:

$$E_{gain} = R_{savings} + C_{savings} + V_{access} + P_{efficiency} \tag{10}$$

Where E_{gain} is the total efficiency gain, $R_{savings}$ is remittance fee savings, $C_{savings}$ is change-related savings, V_{access} is the value of improved vendor access to foreign currency, and $P_{efficiency}$ is price-setting efficiency gain.

9 Conclusion

The Jangara banking system represents an innovative approach to addressing financial challenges faced in both distressed and stable economies. By providing a digital equivalent to the US dollar without the scarcity constraints of physical currency, it offers a practical solution that benefits users across the global financial ecosystem while maintaining a sustainable business model.

The system's unique value proposition lies in its ability to optimize currency flows, ensuring that hard currency reaches those who need it most while providing efficient digital transactions for everyday commerce. By functioning as both a local solution and global framework, Jangara creates a network that grows more valuable with each additional participant, currency, and country.

Mathematical modeling and comparative analysis support the viability of this approach, with significant potential for economic impact in both distressed and stable economies (Raskin and Yermack, 2019; Adrian and Mancini-Griffoli, 2019).

10 Future Work

- Pilot testing in both distressed and stable economies
- Regulatory framework development for cross-border operations
- Partnership with local businesses and international remittance providers
- Development of additional financial services

- Integration with existing payment infrastructure in stable economies
- Research on optimal reserve management across multiple currencies

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